Simulation Project

ETIN71 – Introduction to Integrated Electronics (2026/27)

Topic

Design and test a differential amplifier in Keysight ADS.

Target

Use the design methods and building blocks from the course to construct a differential amplifier.

Getting Started

Install Qucs on your system following the guidelines mentioned in the Qucs-S start-up assistance (separate document). There are separate guidelines if you want to work on a Windows system instead. **Go through the steps highlighted in red to complete the installation**(The other steps are required if you are working on a different Linux system not connected to the EIT server). Create a ETIN70 folder in your home directory for saving your work. Also unzip the Nanoelectronics Student Component Library (NSCL) Qucs Models and add it to your work folder. All the documents and files can be found on Canvas.

First, complete the laboratory experiments in the course, then start working on this project. Create a 'Project' folder for saving your project-related tasks.

Instructions

Your project results must be handed in by Tuesday, October 13, 2026, at 23:59 through a PDF upload in LU Canvas (https://canvas.education.lu.se/) > ETIN71. All parts of the solution must be clearly stated.

The solution must contain:

- Cover page, course code, title, student name(s) in page header; page number in footnote
- Answers to all tasks on the following pages
- Your solution must be edited on a computer

You can discuss with the supervisor at scheduled project supervision sessions. You are required to work in pairs (maximum 2 students per group), unless there is room for individual work in the room.

Work independently and discuss your progress/ challenges during project supervision sessions:

- Carefully read the instructions
- Work actively on the tasks

- Tell the supervisor if you need assistance
- Show your progress to the supervisor

Note

The final course certificate requires approved hand-in assignments, passed laboratory experiments, and a passed final written examination.

Preparation – Nanowire (NW) MOSFET Technology

You will use complementary nanowire (NW) MOSFETs to design a differential amplifier circuit in this project. The tasks will gradually become clearer as the course moves onward. Work continuously.

Open the Nanoelectronics Student Component Library (NSCL) to explore the nanowire (NW) complementary MOS (CMOS) technology. This can be done through the existing NSCL test benches that use transistors in standard size. Don't create new schematics in the NSCL folder.

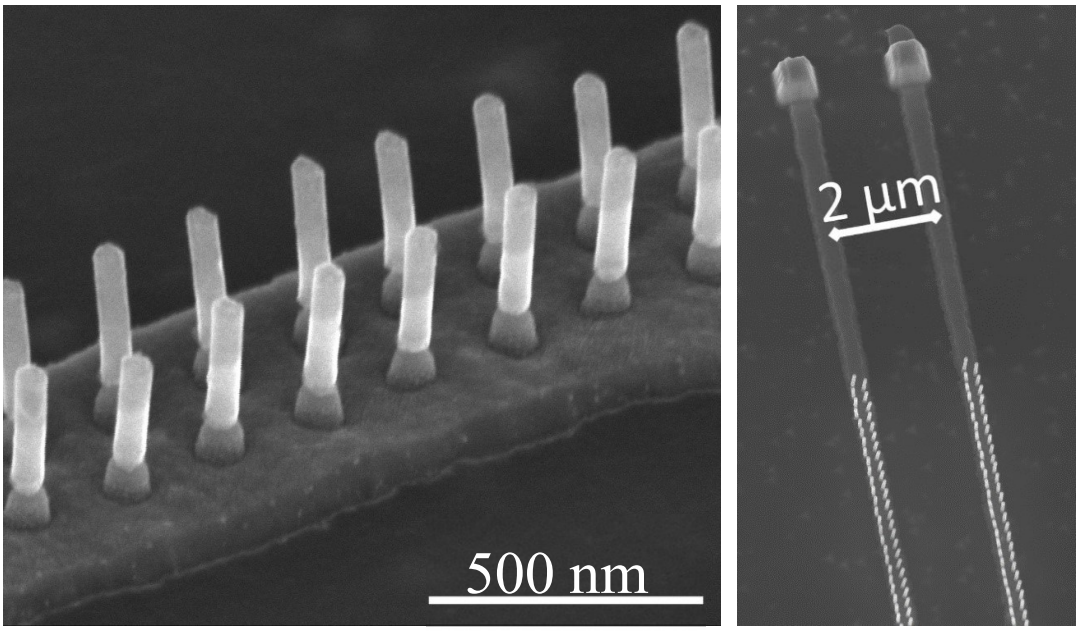
Note that the size of a NW MOSFET is described by the gate length, L, and the number of nanowires, NW. The equivalent gate width per nanowire, W_{eq} = 100 nm, can be used to find the traditional gate width, $W = W_{eq} \cdot NW$, of the NW MOSFET. Also, the parameter number of gate fingers, NF, is available to adapt the geometry of the device. The NWs are typically arranged in multiple parallelconnected gate fingers to reduce series-parasitic resistances.

The threshold voltages of the NW MOSFETs in the NSCL are equal, $V_{tn} = |V_{tp}|$ = 0.15 V. The process transconductance parameters are k'_n = 0.84 S/V for the n-MOS and $k'_p = k'_n/4$ for the p-MOS. Note that these are approximate values since both depend on the operation point of the transistor.

Preparation Task: Characterise the NSCL MOSFETs

Trace and analyse transfer (i_D vs. v_{GS}) and output (i_D vs. v_{DS}) characteristics of the NCSL n- and p-channel MOSFETs, with default parameter values, using the existing pre-defined test benches. **Go through the steps highlighted in pink in the Qucs startup assistance document to complete these tests**

- Simulate and use markers to find the approximate drain current, transconductance, and output resistance of the NSCL_NW_NMOS at V_{GS} = 0.6 V and V_{DS} = 0.6 V
- Simulate and use markers to find the approximate drain current, transconductance, and output resistance of the NSCL_NW_PMOS at V_{GS} = -0.6 V and V_{DS} = -0.6 V



Above: Electron micrograph of a vertical NW n-MOSFET mid-process, after gate definition in Lund NanoLab, adapted from M.E. Sandberg.

Task: Design and Test a Differential Amplifier

Design a two-stage amplifier circuit in the nanowire (NW) complementary MOS (CMOS) technology available from the Nanoelectronics Student Component Library (NSCL). Also, evaluate it as an operational amplifier (op amp) and compare the results to that from an ideal op amp.

The circuit schematic or the differential amplifier is shown in Figure 1 (below) and specific data on the various transistors are given in Table I. If you see a question mark (?) in the specification, the value should be found in the tasks.

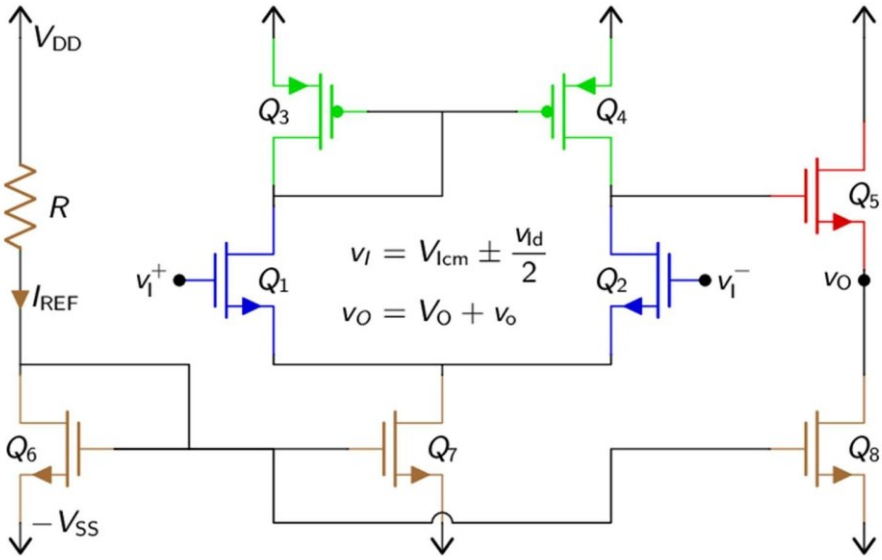


Figure 1: Schematic circuit of the differential amplifier.

The amplifier utilises an NMOS differential pair input stage (Q1-2) loaded by a PMOS current mirror (Q3-4) that feeds the signal to a common drain (CD) output stage (Q5). A current steering circuit is used to set bias currents in the amplifier stages. This analogue circuit utilises dual drive voltages that are symmetric about ground, $V_{DD} = V_{SS} = 1$ V. Note that the inputs and the output terminals carry both a quiescent operating point and a small signal component.

Table I: Specification of the transistors in the differential amplifier.

Transistor	Type	NF (nbr of gate fingers)	L (gate length)	NW (nbr of NW)
Q1	n-MOS	1	20 nm	50
Q2	n-MOS	1	20 nm	50
Q3	p-MOS	1	20 nm	?
Q4	p-MOS	1	20 nm	?
Q5	n-MOS	1	20 nm	?
Q6	n-MOS	1	50 nm	20
Q7	n-MOS	1	50 nm	?
Q8	n-MOS	1	50 nm	?

Detailed Instructions – Working Manual

Build the differential amplifier circuit based on Figure 1, starting with component default sizes. Place pins for inputs, output, and drive voltages, then create a symbol. Create test benches similar to those used in the laboratory experiments (op amp follower) to evaluate if the differential amplifier can perform as an op-amp. You will need DC or transient simulations to set up the circuit and finally make AC simulations to evaluate the functionality of the differential amplifier – compare it to an ideal op-amp.

The following list of tasks can be solved consecutively, but they will require some iterations if you aim for a perfect design. Focus on approximate solution methods, not exact results.

1. Temporarily use a current source to generate the reference current $I_{REF} = 1$ mA
2. Set the size of Q7-8 to ideally yield approximately 10 mA drain current in Q1-5
3. Set the size of Q3-4 to ideally yield similar drain-source voltage over all of Q1-4
4. Adjust the width (integer number of wires) of Q5 to match V_O with V_{Icm} (DC test bench)
5. Replace the reference current source with a resistor

6. Evaluate the amplifier in follower configuration from 20 Hz to 100 GHz (AC test bench)

Report

You will hand in a report to show that you have performed the project tasks and understood the results. Hand in one report per group. The reports will be graded pass or fail, with the opportunity to make one (1) single correction if not passed in the first round.

Hand in the report in PDF format via the corresponding assignment in LU Canvas. Hints for the report structure and content are given below.

COVER PAGE

Title, course code and name of course, full names of all students in the group, date

INTRODUCTION

Goal is to try circuit design in Qucs using NW MOSFET models…

THEORY

Brief descriptions of ideal op amp characteristics, voltage follower configuration, MOSFET low frequency model with transconductance and output resistance, current steering, MOS differential pair, current mirror load, output stage…

METHOD

Explain that you performed and analysed Qucs simulations using a workspace, component library, schematic simulations, data display…

RESULTS &DICUSSION

Show results and discuss the following…

a. Parameters of Q1 n-MOS and Q3 p-MOS low-frequency small signal model (g_m and r_o)
b. Describe solutions to the project tasks (numbered list above)
c. Compare the differential amplifier to an ideal op amp – open loop and voltage follower
d. Discuss flaws of the designed amplifier compared to an ideal op amp (gain, offset, BW, …)